

The extra $(x + 1)$ factor: a proof of Karttunen's second conjecture on the binary columns of the Fibonacci numbers

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Abstract

OEIS A036284 reads one period of the n -th binary column of the Fibonacci numbers as a polynomial over $\text{GF}(2)$. A. Karttunen conjectured in November 1998 that this polynomial is divisible by $(x^3 + 1)^{2^{n-1}-1}$ (Conjecture 1) and that “there is also one extra $(x^1 + 1)$ factor present” (Conjecture 2). We prove Conjecture 2, i.e. that $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ divides the n -th term. Equivalently, the $(x + 1)$ -adic valuation of the column polynomial is at least 2^{n-1} , one more than the valuation of $(x^3 + 1)^{2^{n-1}-1}$. The proof is the same carry-periodicity identity that settles Conjecture 1, read on the factor $x + 1$ instead of on $x^2 + x + 1$; the point is simply that $x + 1$ does not divide $x^2 + x + 1$, so the cancellation that costs one power on the $x^2 + x + 1$ side costs nothing here. The note is self-contained: the identity is restated and reproved in Section 3.

1 The conjecture

Let $F(0) = 0$, $F(1) = 1$, $F(k) = F(k - 1) + F(k - 2)$. The bits in the n -th column from the right of the binary expansions of $F(0), F(1), F(2), \dots$ repeat with period $3 \cdot 2^n$, and OEIS A036284 (A. Karttunen, 1 November 1998, *Periodic vertical binary vectors of Fibonacci numbers*) records one period read as a binary number:

$$a(n) = \sum_{k=0}^{3 \cdot 2^n - 1} \left(\lfloor F(k)/2^n \rfloor \bmod 2 \right) 2^k, \quad a(0), \dots, a(3) = 6, 24, 1440, 5728448.$$

Identifying an integer with the $\text{GF}(2)$ polynomial whose coefficient of x^k is bit k , the entry carries

Conjecture: For $n \geq 1$, each term $a(n)$, when considered as a $\text{GF}(2)[X]$ -polynomial, is divisible by the $\text{GF}(2)[X]$ -polynomial $(x^3 + 1)^{A000225(n-1)}$
Conjecture 2: there is also one extra $(x^1 + 1)$ factor present, see **A136384**.

with $A000225(n - 1) = 2^{n-1} - 1$. Conjecture 1 is proved in a companion note. Here we prove Conjecture 2, in the form

$$(x^3 + 1)^{2^{n-1}-1} (x + 1) \mid a(n) \quad (n \geq 1).$$

As of the “Last modified” line on the live entry (23 August 2026) both statements are still recorded as unproven conjectures and no proof appears in the entry’s links or programs.

2 Notation

Fix $n \geq 1$ and set

$$L = 3 \cdot 2^n, \quad q(x) = x^2 + x + 1, \quad x^3 + 1 = (x + 1)q(x).$$

Note that $q(1) = 1$, so $x + 1 \nmid q$; this small fact is the whole difference between the two conjectures. For nonzero $P \in \text{GF}(2)[x]$ and irreducible f , let $v_f(P)$ be the exponent of f in P . By the Frobenius endomorphism,

$$x^{3 \cdot 2^m} + 1 = (x^3 + 1)^{2^m} \quad (m \geq 0). \quad (1)$$

Let $r_k = F(k) \bmod 2^{n+1}$, let $b(k) = \lfloor r_k/2^n \rfloor \in \{0, 1\}$ be the n -th bit and $s_k = r_k \bmod 2^n$ the low part, so $r_k = b(k)2^n + s_k$. Let

$$\gamma(k) = \lfloor (s_{k-1} + s_{k-2})/2^n \rfloor \in \{0, 1\}$$

be the carry into position n when $F(k-1)$ and $F(k-2)$ are added, and put

$$A(x) = \sum_{k=0}^{L-1} b(k)x^k, \quad \widehat{C}(x) = \sum_{k=0}^{L/2-1} \gamma(k)x^k.$$

A is the polynomial attached to $a(n)$.

3 The identity

Lemma 1. *For $m \geq 1$ the sequence $(F(k) \bmod 2^m)_{k \geq 0}$ is purely periodic with least period $3 \cdot 2^{m-1}$.*

Proof. Pure periodicity holds because $F(k-2) = F(k) - F(k-1)$ makes the recurrence invertible modulo any m ; the value $\pi(2^m) = 3 \cdot 2^{m-1}$ of the Pisano period is classical, see Wall [1]. $\square$

Thus $(r_k)_k$ has period L , so b, s, γ are functions on $\mathbb{Z}/L\mathbb{Z}$ and $r_k \equiv r_{k-1} + r_{k-2} \pmod{2^{n+1}}$ holds for every $k \in \mathbb{Z}/L\mathbb{Z}$.

Lemma 2. *In $\text{GF}(2)[x]$ there is an M with*

$$q(x)A(x) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}} + M(x)(x^3 + 1)^{2^n}. \quad (2)$$

Proof. Splitting each summand into high bit and low part,

$$r_{k-1} + r_{k-2} = (b(k-1) + b(k-2) + \gamma(k))2^n + ((s_{k-1} + s_{k-2}) \bmod 2^n),$$

which is legitimate since $s_{k-1} + s_{k-2} < 2^{n+1}$. Reducing modulo 2^{n+1} and comparing with $r_k = b(k)2^n + s_k$, where $s_k < 2^n$, gives

$$b(k) \equiv b(k-1) + b(k-2) + \gamma(k) \pmod{2} \quad (k \in \mathbb{Z}/L\mathbb{Z}).$$

Multiplying by x^k and summing over $\mathbb{Z}/L\mathbb{Z}$, where the shift $k \mapsto k-1$ becomes multiplication by x modulo $x^L + 1$, yields $q(x)A(x) \equiv \Gamma(x) \pmod{x^L + 1}$ with $\Gamma(x) = \sum_{k=0}^{L-1} \gamma(k)x^k$.

Now $\gamma(k)$ is a function of s_{k-1}, s_{k-2} alone, and $s_j = F(j) \pmod{2^n}$ has period $3 \cdot 2^{n-1} = L/2$ by Lemma 1 applied with $m = n$. So the coefficient list of Γ is two copies of that of $\widehat{C}$:

$$\Gamma(x) = \widehat{C}(x)(1 + x^{L/2}) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}}$$

by (1). Since $x^L + 1 = (x^3 + 1)^{2^n}$, also by (1), this is (2). $\square$

4 The theorem

Theorem 3. *For every $n \geq 1$, $v_{x+1}(A) \geq 2^{n-1}$. Consequently $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ divides A .*

Proof. Both terms on the right of (2) are divisible by $(x^3 + 1)^{2^{n-1}}$ and $v_{x+1}((x^3 + 1)^m) = m$, so $v_{x+1}(qA) \geq 2^{n-1}$. Since $q(1) = 1$ we have $v_{x+1}(q) = 0$, hence $v_{x+1}(A) = v_{x+1}(qA) \geq 2^{n-1}$.

The same equation read with v_q gives $v_q(qA) \geq 2^{n-1}$, and $v_q(q) = 1$, so $v_q(A) \geq 2^{n-1} - 1$ (this is Conjecture 1). Writing $(x^3 + 1)^{2^{n-1}-1}(x + 1) = (x + 1)^{2^{n-1}} q^{2^{n-1}-1}$ and comparing exponents, $v_{x+1}(A) \geq 2^{n-1}$ and $v_q(A) \geq 2^{n-1} - 1$ are exactly what is needed. $\square$

Corollary 4. *For every $n \geq 1$, $a(n)$ is divisible in $\text{GF}(2)[X]$ by $(x^3 + 1)^{2^{n-1}-1}(x + 1)$. This is Karttunen's Conjecture 2.*

Remark (honest accounting). This note and its companion share Lemma 2; once that identity is in hand, Conjecture 1 and Conjecture 2 are the two valuations one can read off it, and Conjecture 2 costs one extra line. They are recorded separately because the entry states them as two separate conjectures, not because the second required new work.

Remark (sharpness). Equation (2) gives the exact value $v_{x+1}(A) = 2^{n-1} + v_{x+1}(\widehat{C})$ whenever $v_{x+1}(\widehat{C}) < 2^{n-1}$. Computation gives $v_{x+1}(\widehat{C}) = 0$ for $n = 1$ and for $4 \leq n \leq 12$, so the bound of Theorem 3 is attained there; at $n = 2$ and $n = 3$ one has $v_{x+1}(\widehat{C}) = 1$ and 3, so $v_{x+1}(A) = 3$ and 7 rather than 2 and 4. Whether $v_{x+1}(\widehat{C}) = 0$ for all $n \geq 4$ — equivalently whether the number of carries in one period of length $L/2$ is odd — is not proved here.

5 Verification

A verification script, written separately from this note, checks every assertion and prints every number quoted. It (i) recomputes $a(0), \dots, a(3)$ and matches the published 6, 24, 1440, 5728448; (ii) checks the carry identity of Lemma 2 at all L residues for $n = 0, \dots, 16$, i.e. up to $L = 196608$, with no failure; (iii) checks the half-period property of γ over the same range, with no failure; (iv) verifies (2) by explicit polynomial division for $n = 1, \dots, 12$; (v) computes $v_{x+1}(A)$, obtaining 2^{n-1} for $n = 1$ and $4 \leq n \leq 12$ and the exceptional 3 and 7 at $n = 2, 3$, together with $v_{x+1}(\widehat{C}) = 0, 1, 3, 0, \dots, 0$; and (vi) divides $a(n)$ by $(x^3 + 1)^{2^{n-1}-1}(x + 1)$ directly for $n = 1, \dots, 12$, obtaining remainder zero in every case.

References

- [1] D. D. Wall, *Fibonacci series modulo m*, Amer. Math. Monthly **67** (1960), 525–532.
- [2] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.
- [3] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A036284>, entry A036284, consulted 23 August 2026.

A Suggested OEIS comment (plain text, house style)

Conjecture 2 is true, by the same identity that settles the first conjecture. Fix $n \geq 1$ and put $L = 3 \cdot 2^n$. With $b(k) = \text{bit } n \text{ of } F(k)$, $A(x) = \sum_{k < L} b(k) x^k$, and $g(k)$ the carry into bit n when $F(k-1)$ and $F(k-2)$ are added, one has $b(k) = b(k-1) + b(k-2) + g(k) \pmod{2}$ for all k in $\mathbb{Z}/L$. Summing x^k over $\mathbb{Z}/L$ gives $(x^{2+x+1}) A(x) = G(x) \pmod{x^L + 1}$. The carry $g(k)$ depends only on $F(k-1)$ and $F(k-2) \pmod{2^n}$, whose period is $L/2$, so $G(x) = C(x)(1 + x^{L/2})$, and in characteristic 2 we have $1 + x^{L/2} = (x^{3+1})^{2^{n-1}}$ and $x^L + 1 = (x^{3+1})^{2^n}$. Hence

$$(x^{2+x+1}) A(x) = C(x) (x^{3+1})^{2^{n-1}} + M(x) (x^{3+1})^{2^n},$$

so both sides are divisible by $(x^{3+1})^{2^{n-1}}$. Since $x+1$ does not divide x^{2+x+1} , no power of $x+1$ is lost on the left: the $(x+1)$ -multiplicity of $a(n)$ is at least 2^{n-1} , whereas $(x^{3+1})^{2^{n-1}-1}$ only accounts for $2^{n-1} - 1$ of them. That is the extra $(x+1)$ factor. The count is exactly 2^{n-1} for $n = 1$ and $n \geq 4$, and $2^{n-1} + 1$, $2^{n-1} + 3$ at $n = 2, 3$.